# Employee Management Database Design

## Project Overview

The Employee Management System is designed to manage employees, departments, projects, user accounts, and roles within an organization.

The database is built using PostgreSQL and follows relational database design principles including normalization, primary keys, foreign keys, and many-to-many relationships.

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## Business Requirements

The system should support:

- Managing employee information.
- Managing departments.
- Assigning employees to departments.
- Managing projects.
- Assigning employees to multiple projects.
- Managing user accounts and roles.

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## Database Tables

### Department

Stores department information.

| Column | Type | Description |
|---|---|---|
| department_id | SERIAL | Primary Key |
| department_name | VARCHAR(100) | Department Name |
| location | VARCHAR(100) | Department Location |

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### Employee

Stores employee details.

| Column | Type | Description |
|---|---|---|
| employee_id | SERIAL | Primary Key |
| first_name | VARCHAR(50) | Employee First Name |
| last_name | VARCHAR(50) | Employee Last Name |
| email | VARCHAR(100) | Unique Email |
| salary | NUMERIC(10,2) | Employee Salary |
| department_id | INT | Foreign Key |

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### Project

Stores project details.

| Column | Type | Description |
|---|---|---|
| project_id | SERIAL | Primary Key |
| project_name | VARCHAR(100) | Project Name |
| start_date | DATE | Start Date |
| end_date | DATE | End Date |

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### Employee_Project

Junction table used for many-to-many relationship.

| Column | Type |
|---|---|
| employee_id | INT |
| project_id | INT |

Composite Primary Key:

(employee_id, project_id)

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### Role

Stores system roles.

Examples:

- Admin
- Manager
- Employee

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### User_Account

Stores login information.

Each user is associated with one role.

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## Relationships

### One-to-Many

Department → Employee

One department can have many employees.

Employee → Department

Many employees belong to one department.

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### Many-to-Many

Employee ↔ Project

One employee can work on multiple projects.

One project can have multiple employees.

Implemented using Employee_Project table.

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## Entity Relationship Diagram (ERD)

Department (1) | | v Employee (M)

Employee (M) | | v Employee_Project ^ | | Project (M)

Role (1) | | v User_Account (M)

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## Normalization

The database follows Third Normal Form (3NF):

### 1NF

- No repeating groups.
- Atomic column values.

### 2NF

- No partial dependency.

### 3NF

- No transitive dependency.

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## Constraints Used

### Primary Keys

- department_id
- employee_id
- project_id
- role_id
- user_id

### Foreign Keys

- employee.department_id
- employee_project.employee_id
- employee_project.project_id
- user_account.role_id

### Unique Constraints

- employee.email
- user_account.username

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## Indexing Strategy

Index created on:

- employee.email

Purpose:

- Faster employee lookup.
- Improved login and search performance.

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## Future Enhancements

- Attendance Management
- Leave Management
- Salary History
- Audit Logs
- Manager Hierarchy
- REST API Integration using Spring Boot
- JWT Authentication
- Reporting Dashboard

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## Technologies Used

- PostgreSQL
- SQL
- pgAdmin
- Git
- GitHub